TrustiPay : AI Powered, Bilingual and Offline Enabled Digital Payment Platform

Saparamadu D.D.D.T
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
devramsaparamadu@gmail.com

Weligamage C.N
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
chinthanaweligamage1@gmail.com

Kiriella K.A.D.N
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
dilmanikiriella@gmail.com

Nadeeshani K.M.I
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
imeshanadeeshani761@gmail.com

Anjalie Gamage
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
anjalie.g@sliit.lk

Dhammika De Silva
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
dhammika.d@sliit.lk

Abstract*— Significant advancements have been made in the field of digital payment infrastructure in developing countries. However, the use of digital payment solutions is hindered by a range of obstacles beyond those related to speed of transaction processing. In particular, accessibility challenges, fraud exposure, unstable connectivity, and the limited decision value of raw transaction records continue to affect user trust and everyday usability. This paper presents TrustiPay, a prototype digital payment platform that addresses these challenges through a unified system architecture in which interaction, risk assessment, transaction management, and insight are handled in a coherent way within a single lifecycle process. To maintain consistency under unstable connectivity, transactions are processed through peer-to-peer mechanisms which utilizes tamper-evident store-and-forward model with integrity validation, duplicate detection, and double-spend prevention. Synchronized transactions undergo deferred fraud assessment prior to final confirmation, ensuring that offline continuity does not bypass security controls. Risk evaluation is performed using a hybrid approach that combines anomaly detection, supervised learning, and contextual behavioral features to enable adaptive decisioning. In addition, transaction histories are transformed into interpretable financial indicators, behavioral profiles, and personalized user guidance, extending the system beyond transaction execution to decision support. The prototype is realized through an Android client and FastAPI-based fraud, ledger, and analytics backends. Instead of relying on unsupported benchmark claims, the paper contributes clear architecture, implementation-driven design rationale, and prototype-oriented evaluation of an inclusive and security-aware payment system. The results suggest that accessibility, adaptive security, offline continuity, and financial guidance should be treated as interdependent design requirements in next-generation digital payment platforms.).

***Keywords*— digital payment systems, fraud detection, offline transactions, peer-to-peer payments, anomaly detection, behavioral analytics, financial decision support, mobile payment platforms, distributed transaction systems**

I. Introduction

Sri Lanka's fast-payment infrastructure has expanded rapidly in recent years, with CEFTS becoming an important interoperability rail for retail fund transfers. Its share of interbank retail transactions increased from 8% in 2018 to 67% in 2024, while annual throughput reached 203 million transactions in 2024 [1], [2]. However, the growth of payment infrastructure does not automatically ensure routine, inclusive, and confident use. Sri Lanka already has broad mobile broadband coverage, with 3G and 4G coverage reaching 95% and 96% respectively, yet only 51% of individuals actively use the internet and only 55% have made or received a digital payment [1]. This suggests that a payment platform may be technically efficient at the network level while still remaining difficult to trust and use in everyday settings. In payment systems, trust extends beyond fraud prevention to include interaction clarity, recoverability, and predictable system behavior under uncertainty. This broader perspective is particularly important in emerging-economy contexts, where accessibility barriers, fraud exposure, and connectivity instability often appear together. Previous studies have demonstrated that even with mobile banking applications, visual impairment and aging can pose considerable challenges, particularly if interface design lacks inclusiveness and the user trust is low[3], [4]. With respect to Sri Lanka, this becomes even more relevant with the latest finding indicating that while 82% of older individuals engage with digital technology, usage drops off considerably with age, especially with individuals between 75 and 80 years old [17].These patterns indicate that digital access alone is not enough; usability, confidence, and local context remain equally important.

Although prior research has explored accessible mobile banking, fraud detection, offline payments, and transaction based financial guidance, these areas are typically addressed as separate streams [8]-[16]. This creates an important gap at the architectural level. A voice interface may improve convenience, but if it is weakly bound it can also introduce risk. Fraud controls that are too rigid may disrupt legitimate users. Offline continuity without strong integrity checks can create opportunities for misuse. Similarly, analytics that merely summarize past spending may offer limited practical value. The issue addressed in this work is therefore not the absence of solutions in individual subdomains, but the lack of an integrated payment architecture in which inclusion, adaptive security, resilience, and financial guidance work together.

To address this gap, this paper presents TrustiPay, a prototype digital payment platform designed for emerging economic conditions. TrustiPay brings together guided interaction, risk-adaptive fraud control, resilient offline value transfer, and interpretable financial analytics within a single architecture. Rather than treating these capabilities as separate add-on features, the platform is designed on the premise that they must operate together if digital payments are to become both usable and trustworthy. The contributions of this work are threefold. First, it presents a unified payment architecture that couples accessibility, adaptive security, resilient transaction continuity, and financial guidance around one shared transaction lifecycle. Second, it defines an execution model in which online transactions remain safety-bounded, offline transfers remain auditable and fraud-reviewable after synchronization, and transaction histories remain available for interpretable post-transaction guidance. Third, it demonstrates, through prototype oriented evaluation, that these requirements can be implemented as one coordinated system without relying on unsupported large-scale benchmark claims.

II. Related Work

A. Digital Payment Options and Accessibility

Digital payment systems have significantly expanded financial inclusion by using mobile and web-based technology to manage transactions in an easy, efficient, and scalable manner [18], [19].

However, existing voice- enabled digital financial systems lack the integration of safety-bounded execution mechanisms. They allow direct action triggering without the presence of adequate validation mechanisms. Moreover, the integration of multilingual support for low-resource languages such as the Sinhala language remains considerably limited due to the challenges in speech recognition accuracy and intent disambiguation [21]. Existing systems improve the accessibility of financial services but lack the required safety and robustness.

B. Financial Transaction Fraud Detection

Financial systems have made extensive use of machine learning-based fraud detection systems, which include hybrid techniques, anomaly detection, and supervised classification to find suspicious transactions [22], [23]. These algorithms forecast the likelihood of fraudulent financial transactions using a variety of characteristics, including transaction volumes and frequencies.

Although machine learning-based fraud detection systems have been seen to be successful in detecting fraudulent financial transactions, they face challenges such as class imbalance and concept drift, and high false-positive rates [23]. Moreover, most fraud detection models operate as standalone classifiers and are not tightly integrated with transaction execution pipelines and often lack the integration of contextual enrichment mechanisms such as behavioral user profiles and event-based signals.

C. Offline Payment and Decentralized Transaction Systems

Offline digital payment systems have been proposed to address the limitations in connectivity in emerging markets. These systems allow transactions to be processed using device-level processing and delayed synchronization [24], [25]. Techniques such as local ledger, peer-to-peer communication, and store and forward transmission have been used to allow the continuity of transactions in offline environments.

However, such systems also face the challenge of addressing the limitations in the ordering, replay resistance, and the double-spend prevention in transactions[25]. Most existing approaches primarily focus on the resilience of the system in the case of connectivity but lack integrated fraud detection and consistent reconciliation mechanisms. Additionally, the methods used in the propagation of transactions are often not designed for efficient dissemination under intermittent connectivity, limiting reliability in real-world deployment scenarios

D. Financial Behavior Analysis and Decision Support

Financial analytics systems have traditionally been based on the tracking ,visualization and providing users with expense summaries and category-based insights [26]. However, more recent approaches to financial analytics have been based on the use of behavioral analysis techniques such as anomaly detection and clustering to identify spending patterns and user profiles [27].

Nevertheless, most financial analytics approaches provide the user with raw data and fragmented information without actionable guidance while lacking interpretable composite metrics that quantify overall financial health. Additionally, existing financial analytics approaches rarely integrate financial analytics with transaction systems, limiting their ability to provide real-time decision support or influence user behavior effectively.

E. Research Gap and Contribution

From the above discussion, it is clear that existing research has been based on the provision of accessibility, fraud detection, offline transactions, and financial analytics. However, limited work has explored their integration within a unified payment architecture that coordinates these functionalities across a single transaction lifecycle.

Unlike previous studies, in which the concentration focused on individual features, the proposed system in TrustiPay is implemented as an integrated system, which brings together bilingual voice-interactivity, risk-adaptive fraud decisioning, robust offline transaction propagation, and interpretable financial analytics. This proposed unified approach ensures secure, accessible, and adaptive digital payments while maintaining consistency across both online and offline environments.

III. Methodology

A. Design Objective and Integrated Architecture

TrustiPay is developed using a prototype-oriented systems methodology in which the research objective is addressed through the design and implementation of an integrated digital payment architecture. The system is designed to evaluate whether accessibility, adaptive fraud control, resilient transaction continuity, and behavioral financial analytics can operate cohesively within a single end-to-end platform. Rather than treating these capabilities as independent modules, TrustiPay organizes them around one transaction lifecycle in which user interaction, execution control, risk evaluation, synchronization, and guidance remain coordinated. Fig. 1 illustrates the transaction lifecycle within the TrustiPay system.

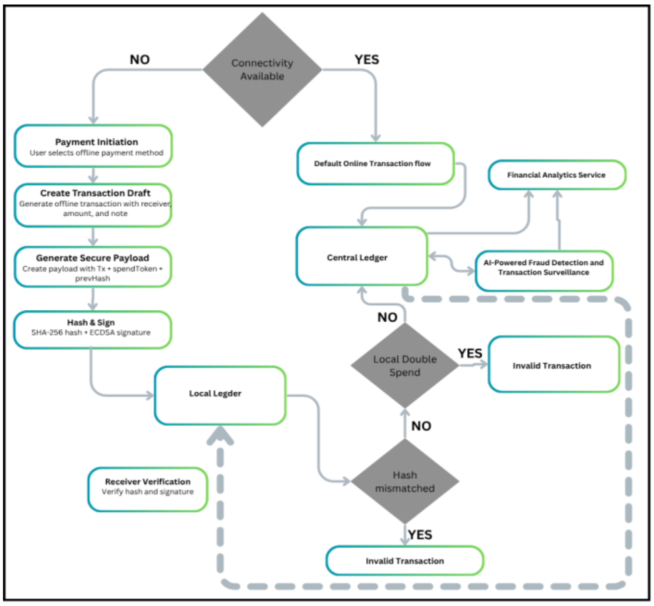

Fig. 1. Transaction lifecycle

The architecture consists of a mobile client and three backend service domains: fraud and risk evaluation, transaction orchestration with offline support, and financial analytics. The client supports wallet interaction, transaction confirmation, voice-assisted guidance for the online path, and temporary local storage for disconnected events. The backend is implemented using a modular FastAPI-based design so that risk scoring, ledger synchronization, and analytics can evolve independently while still participating in the same execution flow. Fig. 2 summarizes the overall system architecture.

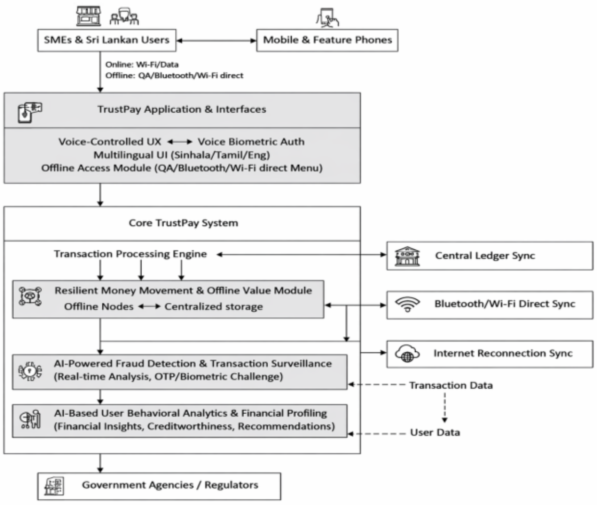

Fig. 2. Overall System Architecture

B. *Unified Transaction Lifecycle and Risk-Aware Online Execution*

Transaction processing begins at the client layer. In the online path, bilingual voice assistance is used only as a guidance mechanism; it can support navigation and parameter prefilling, but it cannot directly commit financial actions. High-confidence navigation and parameter-prefill intents are accepted, while sensitive actions are always routed to explicit confirmation and ambiguous input triggers retry. This preserves accessibility gains without weakening transaction control. Once a transaction is ready for submission, it is evaluated by a hybrid fraud decision service. For a transaction feature vector x, an anomaly score A(x) and a supervised fraud probability C(x) are computed and combined as shown below:

$$R(x) = \lambda A(x) + (1 - \lambda)C(x), 0 \leqslant R(x) \leqslant 1 \qquad (1)$$

The Augmentation of risk score relies on contextual signals derived from behavioral profiles and event-level features, including device familiarity, location novelty, transaction velocity, and network interactions. A policy layer then converts the combined risk state into one of three adaptive outcomes: low-risk transactions proceed with minimal friction, medium-risk transactions are escalated through step-up verification, and high-risk transactions are blocked and flagged for investigation. In this way, TrustiPay treats fraud handling as transaction decisioning rather than as standalone classification.

TABLE I. TRANSACTION STATES ACROSS ONLINE AND OFFLINE EXECUTION

Path	Fraud evaluation	Finalization
Guided online transaction	Immediate, before commit	Outcome returned directly by policy service
Disconnected offline transfer	Deferred, after synchronization	No final confirmation until fraud review is complete

C. *Resilient Offline Execution and Deferred Fraud Confirmation*

When connectivity is unavailable, TrustiPay shifts to an offline-first execution model in which client devices act as autonomous nodes capable of generating, validating, and storing transactions locally. Offline exchanges are carried over peer-to-peer channels such as QR, Wi-Fi Direct, and Bluetooth, and local dissemination is strengthened through a gossip-inspired relay strategy that propagates only missing records under bounded hop and time-to-live constraints. This design supports resilient money movement under intermittent connectivity rather than treating disconnection as a simple temporary queue.

The construction of each transaction uses device-specific unique identifier and is linked to prior state through a tamper-evident ledger. The transaction identifier is derived as:

$$tx\_id = H(device\_id \mathbin{\|} counter \mathbin{\|} timestamp \mathbin{\|} prev\_hash) \qquad (2)$$

where the monotonic device counter promotes ordering and uniqueness. To maintain the integrity of the local ledger, each entry is hashlinked as follows:

$$entry\_hash = H(payload\_hash \mathbin{\|} prev\_hash \mathbin{\|} signature) \qquad (3)$$

ensuring that any changes, additions, or deletions made within the local store are noticeable. After connectivity is restored, offline transactions are synchronized with the central ledger. However, synchronization does not imply immediate finalization. Rather, synced records proceed into a fraud_detection_pending state and are only elevated to confirmed status following the online path's fraud decision pipeline, duplicate handling, counter-based ordering checks, signature verification, and dispute settlement. This preserves consistency between real-time and deferred execution while keeping offline continuity subject to common trust controls.

D. Integrated Behavioral Intelligence and User Guidance

TrustiPay extends beyond transaction execution by reusing normalized transaction history from both online and offline flows as the basis for behavioral intelligence. The analytics service derives features such as savings ratio, nonessential spending proportion, spending stability, transaction frequency, and anomaly incidence. Let z_u denote the behavioral feature vector for user u. The system computes a Financial Health Score (FHS) as a calibrated representation of financial stability:

$$\hat{y}\_u = f\_RF(z\_u), FHS\_u = Cal(\hat{y}\_u, z\_u) [0, 100] \quad (4)$$

where $f\_RF(\cdot)$ denotes the predictive modeling layer and $Cal(\cdot)$ denotes calibration and interpretation. The resulting score is complemented by behavioral clustering and rule-based recommendation generation so that users receive actionable explanations, top contributing factors, and personalized guidance rather than raw ledger summaries alone. In this formulation, analytics is not an isolated afterthought; it is a downstream interpretation layer attached to the same transaction lifecycle that supports payment execution and fraud control.

E. Prototype Implementation and Evaluation Approach

The TrustiPay prototype is implemented using an Android client and FastAPI-based backend services for fraud detection, transaction orchestration, and analytics. Reproducible local deployment is achieved via SQLite, with the modular service structure supports independent testing and system integration. Because institution-scale banking datasets and deployment logs were not available, evaluation is conducted through scenario-based validation using synthetic and developer-generated transaction data, bilingual online interactions, and disconnected synchronization cases. The evaluation therefore focuses on system-level properties, including interaction safety, adaptive risk handling, resilient continuity, and decision-support effectiveness, rather than unsupported large-scale benchmark claims.

IV. RESULTS AND EVALUATION

A. Integrated Transaction Workflow Efficiency

The prototype successfully supported the intended payment flow under both connected and disconnected operating conditions. In the online path, bilingual voice guidance assisted users with navigation and transaction preparation, but sensitive actions still required explicit confirmation before submission. Once submitted, transactions were evaluated by the fraud-decision engine and routed according to risk level: low-risk transactions were completed directly, medium-risk cases were escalated through OTP verification, and high-risk cases were blocked before final commitment. In the offline path, transactions were created and stored locally during connectivity loss, then synchronized and re-evaluated after reconnection before finalization. Analytics outputs were generated only after successful completion, ensuring that financial guidance was based on confirmed transaction history rather than tentative records.

B. Fraud-Decision Performance

The fraud-decision layer was evaluated on the transaction dataset using a stratified split. Because the dataset is highly imbalanced, model performance was assessed with precision, recall, F1-score, ROC-AUC, and PR-AUC in addition to accuracy. Table II compares the proposed fraud pipeline with relevant baseline models. Tree-based ensemble methods consistently outperformed the linear baseline, while the proposed hybrid model, which combines anomaly-aware scoring with supervised classification, achieved strong and balanced performance. In particular, the proposed hybrid reached 98.99% accuracy, 88.38% recall, and 88.52% F1-score. These results indicate that the fraud layer is capable of separating suspicious transactions effectively while remaining suitable for policy-based intervention within the payment flow.

TABLE II. FRAUD-DECISION MODEL COMPARISON ON THE EVALUATION DATASET

Model	Accuracy	Recall	F1-score
Logistic Regression	93.97%	85.62%	1.91%
Balanced Random Forest	95.18%	86.88%	13.28%
Random Forest	97.57%	50.00%	66.67%
Proposed Hybrid (Isolation Forest + Random Forest)	**98.99%**	**88.38%**	**88.52%**

C. Offline Execution and Deferred Confirmation

Scenario-based evaluation also showed that offline transaction handling remained consistent with the intended control logic of the platform. Transactions created during disconnection were not treated as completed at the point of local capture. Instead, they remained pending until synchronization, signature verification, ordering checks, duplicate handling, and fraud review were completed. This prevented the offline path from bypassing the same trust controls applied in the online path. The resulting behavior confirms that offline capability in TrustiPay functions as controlled deferred execution rather than as unrestricted local settlement.

D. Post-Transaction Analytics and Decision Support

The analytics service produced Financial Health Scores, behavior profiles, and recommendation outputs from confirmed transaction histories. This allowed the platform to extend beyond transaction execution and fraud prevention by converting normalized payment records into interpretable financial guidance. In the evaluated scenarios, analytics outputs were generated only after successful completion or reconciliation, which preserved consistency between transaction status and user-facing recommendations. This is important because it ensures that financial guidance reflects finalized system state rather than provisional activity.

E. End-to-End System Validation

Taken together, the evaluation shows that TrustiPay supports the intended payment lifecycle across normal online payments, elevated-risk transactions, blocked suspicious cases, and disconnected transaction scenarios. The strongest quantitative evidence comes from the fraud-decision layer, while the scenario-based assessment shows that the wider system behaves consistently under different operating conditions. The main result is therefore that TrustiPay functions as a workable digital payment system with guided interaction, adaptive risk handling, controlled offline execution, and post-transaction financial guidance within a single operational flow.

V. Discussion

A. Overall Discussion

TrustiPay differs from many conventional payment applications in the way it approaches trust. Instead of treating accessibility, fraud control, offline support, and user guidance as separate features, the platform is designed around the idea that these concerns are closely connected. The voice layer improves accessibility but still keeps sensitive actions under explicit confirmation. The fraud layer introduces graded intervention rather than relying on a simple allow-or-reject model. The offline path supports continuity, while still preserving integrity checks before finalization. The analytics layer adds another dimension by turning transaction histories into understandable financial guidance. Considered together, these components suggest that usability, security, continuity, and decision support do not have to be treated as competing priorities in payment-system design.

B. Security and Privacy Considerations

Security and privacy were considered throughout the TrustiPay design, even at the prototype stage. Voice interaction is limited to guidance and prefilling, while sensitive actions always return to explicit user confirmation. Replay and duplicate risks are reduced through transaction identifiers and idempotency controls, and offline transactions are checked for integrity and consistency before they are finalized. Suspicious cases may be escalated through OTP-based step-up verification. In addition, backend processing is limited to the transaction and profile data required for fraud analysis and analytics generation. Although the system is not presented as a production-certified solution, these design choices show that safety and privacy were treated as core architectural concerns from the beginning.

VI. Conclution

This paper presented TrustiPay, a design and prototype for an inclusive, bilingual, and offline-enabled digital payment platform. The core contribution is architectural: the system integrates safe voice-assisted interaction, adaptive fraud decisioning, offline transaction resilience, and interpretable financial analytics within one modular payment framework. By doing so, it addresses a practical gap in current payment system design, where accessibility, trust, continuity, and user guidance are often treated as separate concerns. The prototype shows that these requirements can be engineered together without relying on unsafe automation or unsupported performance claims. TrustiPay therefore provides a credible foundation for future deployment-oriented research, broader empirical validation, and integration with real payment ecosystems.

VII. Future Work

A. Controlled User Studies for Accessibility and Interaction

A practical next step would be to study the VOINT interaction model with real users, especially visually impaired users, older adults, and bilingual speakers. This would make it possible to assess whether the system improves usability, confidence, and task completion in comparison with conventional screen-based payment flows. Such a study would also help determine whether the current balance between voice assistance and confirmation is appropriate in everyday use.

B. Institution-Scale Fraud Evaluation

The fraud module currently demonstrates a layered decision process, but its performance should eventually be tested on broader and more realistic financial datasets. Future work should therefore include institutionally governed transaction records so that false positives, calibration behavior, drift resilience, and decision reliability can be examined under more demanding conditions. This would provide a stronger basis for judging how the fraud layer performs beyond prototype scenarios.

C. Stronger Offline Security and Device Trust

The offline workflow already includes integrity checks, duplicate protection, and controlled synchronization, but future versions should strengthen trust at the device level as well. Secure key storage, device attestation, stricter replay resistance, and capped offline transaction limits would make the disconnected payment path more robust. These improvements would be particularly important if the platform were extended toward real deployment.

D. Longitudinal Analytics and Personalized Financial Guidance

The current analytics service provides interpretable financial indicators and behavior profiles from transaction histories. Future work should expand this toward longitudinal financial modeling, recommendation tracking, and outcome-based personalization. This would allow the system not only to describe user behavior at a given point in time, but also to measure improvement, detect deterioration, and deliver more context-sensitive financial guidance over time.

References

[1] The World Bank, "Sri Lanka Digital Transformation (P508317): Project Information Document (Concept Stage)," Washington, DC, USA, Jun. 2025.

[2] K. Ambagahawita, "Fast Payments for Everyone - CEFTS Implementation in Sri Lanka," Central Bank Payment News, Issue 8.2, Central Bank of Sri Lanka, Feb. 24, 2025.

[3] Alayed, “Are Saudi Arabian banks’ mobile applications accessible for blind or partially sighted users?: a customers’ perspective and evaluation,” Univ Access

Inf Soc, vol. 24, no. 1, pp. 585–606, Mar. 2025, doi: 10.1007/s10209-023-01082-y.

[4] X. Jin, E. Kuang, and M. Fan, “”Too old to bank digitally? ”: A Survey of Banking Practices and Challenges Among Older Adults in China,” in Designing Interactive Systems Conference 2021, Virtual Event USA: ACM, Jun. 2021, pp. 802–814. doi: 10.1145/3461778.3462127.

[5] B. Gamage, R. Pushpananda, T. Nadungodage, and R. Weerasinghe, “Applicability of End-to-End Deep Neural Architecture to Sinhala Speech Recognition,” Int J on Adv. in ICT for Emerging Countries, vol. 17, no. 1, pp. 17–21, May 2024, doi: 10.4038/icter.v17i1.7273.

[6] B. Atil et al., “Non-Determinism of ‘Deterministic’ LLM Settings,” 2024, arXiv. doi: 10.48550/ARXIV.2408.04667.

[7] J. Wang, Z. Xu, W. Zhai, and F. Xu, “Exploring Confirmation Strategies for Voice Interaction in Multi-Tasking Scenario,” Int J of Soc Robotics, vol. 17, no. 11, pp. 2755–2768, Nov. 2025, doi: 10.1007/s12369-025-01302-w.

[8] A. Compagnino et al., “An Introduction to Machine Learning Methods for Fraud Detection,” Applied Sciences, vol. 15, no. 21, p. 11787, Nov. 2025, doi: 10.3390/app152111787.

[9] N. Baisholan, J. E. Dietz, S. Gnatyuk, M. Turdalyuly, E. T. Matson, and K. Baisholanova, “A Systematic Review of Machine Learning in Credit Card Fraud Detection Under Original Class Imbalance,” Computers, vol. 14, no. 10, p. 437, Oct. 2025, doi: 10.3390/computers14100437.

[10] Vorobyev and A. Krivitskaya, “Reducing false positives in bank anti-fraud systems based on rule induction in distributed tree-based models,” Computers & Security, vol. 120, p. 102786, Sep. 2022, doi: 10.1016/j.cose.2022.102786.

[11] D. Temoshok et al., “NIST SP 800-63-4: digital identity guidelines,” National Institute of Standards and Technology (U.S.), Gaithersburg, MD, NIST SP 800-63-4, Aug. 2025. doi: 10.6028/NIST.SP.800-63-4.

[12] BIS Innovation Hub, Project Polaris: Handbook for Offline Payments with CBDC. Basel, Switzerland: Bank for International Settlements, 2023.

[13] T. Babina et al., “Customer data access and fintech entry: Early evidence from open banking,” Journal of Financial Economics, vol. 169, p. 103950, Jul. 2025, doi: 10.1016/j.jfineco.2024.103950.

[14] E. E. Firat et al., "MoneyVis: Open Bank Transaction Data for Visualization and Beyond," in EuroVis 2023 - Short Papers, 2023.

[15] S. R. Baker and L. Kueng, “Household Financial Transaction Data,” Annu. Rev. Econ., vol. 14, no. 1, pp. 47–67, Aug. 2022, doi: 10.1146/annurev-economics-051520-023646.

[16] Central Bank of Sri Lanka and International Finance Corporation, Financial Literacy Survey Sri Lanka - 2021: Key Survey Findings, Colombo, Sri Lanka, 2022.

[17] N. Sumathipala, D. Y. Abeywardhana, N. Nanayakkara, S. Rasnayake, P. Kirishanthan, S. Yatigammana, R. de Silva, and C. Rughiniș, “Bridging the digital divide: Factors influencing the adoption and use of digital platforms among older adults in Sri Lanka,” Socio-economic and Health Perspectives, 2025.

[18] T. Dahlberg, N. Mallat, J. Ondrus, and A. Zmijewska, “Past, present and future of mobile payments research: A literature review,” Electronic Commerce Research and Applications, vol. 7, no. 2, pp. 165–181, Jun. 2008, doi: 10.1016/j.elerap.2007.02.001.

[19] World Bank, "The Global Findex Database 2021: Financial Inclusion, Digital Payments, and Resilience," World Bank Group, 2021.

[20] A. Khan, M. Akhtar, S. M. Qureshi, M. Mustafa, N. A. Alsaleh, and I. Ahmad, “A Systematic Review of Safety-Driven Approaches in Human–Robot Collaborative Systems,” Sensors, vol. 26, no. 7, p. 2079, Mar. 2026, doi: 10.3390/s26072079.

[21] A. Graves, A. Mohamed, and G. Hinton, “Speech Recognition with Deep Recurrent Neural Networks,” 2013, arXiv. doi: 10.48550/ARXIV.1303.5778.

[22] J. Jurgovsky et al., “Sequence classification for credit-card fraud detection,” Expert Systems with Applications, vol. 100, pp. 234–245, Jun. 2018, doi: 10.1016/j.eswa.2018.01.037.

[23] A. Dal Pozzolo, O. Caelen, Y.-A. Le Borgne, S. Waterschoot, and G. Bontempi, “Learned lessons in credit card fraud detection from a practitioner perspective,” Expert Systems with Applications, vol. 41, no. 10, pp. 4915–4928, Aug. 2014, doi: 10.1016/j.eswa.2014.02.026.

[24] N. Narula, W. Vasquez, and M. Virza, "Offline cryptocurrency payments," MIT Digital Currency Initiative, 2018.

[25] Ali, Muneeb & Nelson, Jude & Shea, Ryan & Labs, Blockstack & Freedman, Michael. (2016). Blockstack: A Global Naming and Storage System Secured by Blockchains.

[26] A. Lusardi and O. S. Mitchell, “The Economic Importance of Financial Literacy: Theory and Evidence,” SSRN Journal, 2013, doi: 10.2139/ssrn.2260193.

[27] Schafer, Ben & Frankowski, Dan & Dan, & Herlocker, & Jon, & Shilad, & Sen, Shilad. (2007). Collaborative Filtering Recommender Systems.